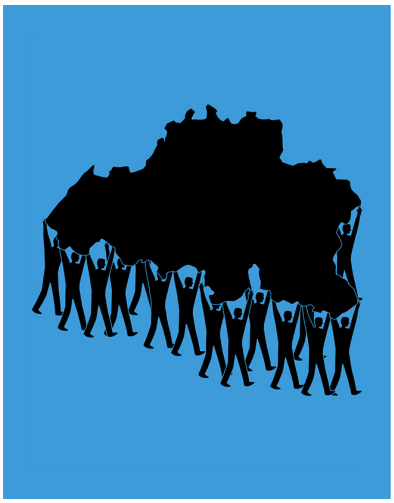


THE VIRTUAL INCREASE OF THE POPULATION IS LIMITED BY THE FERTILITY OF THE COUNTRY

THE VERHULST EQUATION

IN CONTEXT

KEY FIGURES

Thomas Malthus (1766–1834), **Pierre-François Verhulst** (1804–49)

BEFORE

1798 Thomas Malthus argues that populations increase exponentially, based on a common ratio, whereas food supplies grow more slowly at a constant rate, leading to potential food shortages.

1835 Belgian statistician Adolphe Quetelet suggests that population growth tends to slow down as population density increases.

AFTER

1911 Anderson McKendrick, working as an army physician, applies the Verhulst equation to bacteria populations.

1920 American biologist Raymond Pearl proposes the Verhulst equation as a “law” of population growth.

Pierre-François Verhulst was a Belgian mathematician who, after reading Thomas Malthus’s *An Essay on the Principle of Population*, became fascinated by human population growth. In 1845, he published his own model for population dynamics, which was later named the Verhulst equation.